## DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot D

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*Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.*

### Case File CD-D16: The Fitness centre Case [20 marks]

**Briefing:** You are the data detective hired by the owner of a fitness centre. The file holds 14 daily records from the training season, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

| R name | What it measures | R name | What it measures |
|---|---|---|---|
| outdoor_temp | Outdoor temperature (deg C) | injuries | Injury reports |
| gym_visits | Gym visits | trainers | Trainers on duty |
| shakes_sold | Protein shakes sold | towel_loads | Towel laundry loads |
| load_dev | Training load minus optimum (units) | | |

**Evidence: run this code in R to create your case file (do not edit it).**

```
set.seed(758362)
n <- 14
outdoor_temp <- round(rnorm(n, 25.8, 4.7), 1)
gym_visits <- round(189 + 47.1 * (0.94 * (outdoor_temp - 25.8) / 4.7 + 0.35 * rnorm(n)), 0)
shakes_sold <- round(52 + 14.1 * (-0.89 * (outdoor_temp - 25.8) / 4.7 + 0.45 * rnorm(n)), 0)
load_dev <- round(seq(-18.1, 18.1, length.out = n)[order(runif(n))], 1)
injuries <- round(1.7 + 10.0 * (load_dev / 18.1)^2 + rnorm(n, 0, 0.5), 0)
trainers <- round(rnorm(n, 5, 1.4), 0)
towel_loads <- round(15 + 4.7 * (0.24 * (trainers - 5) / 1.4 + 0.97 * rnorm(n)), 0)
trainers[6] <- 10; towel_loads[6] <- 30   # row 6 was typed in by a trainee
case <- data.frame(outdoor_temp, gym_visits, shakes_sold, load_dev, injuries, trainers, towel_loads)
```

**Tasks:**

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot injuries against load_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot towel_loads against trainers. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. gym_visits and shakes_sold are strongly linked. Report r(gym_visits, shakes_sold), then r(outdoor_temp, gym_visits) and r(outdoor_temp, shakes_sold). Compute the partial correlation of gym_visits and shakes_sold controlling for outdoor_temp: correlate resid(lm(gym_visits ~ outdoor_temp)) with resid(lm(shakes_sold ~ outdoor_temp)). Write a two-sentence verdict: does gym_visits cause shakes_sold? Name one study design that could test causation. [5]

***Detective's rule:*** *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

*Code of Conduct: I certify that this work is my own. Signature: _____*

**1.Question:** *Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix.*

## R code

```
cm <- round(cor(case), 2)
print(cm)

n <- nrow(case); n              # 14
choose(7, 2)                    # distinct pairs

# Correlogram (ggplot2 geom_tile + geom_text)
m <- as.data.frame(as.table(cor(case)))
names(m) <- c("V1", "V2", "r")
m$lab <- sprintf("%.2f", m$r)
m$V1 <- factor(m$V1, levels = colnames(case))
m$V2 <- factor(m$V2, levels = rev(colnames(case)))

ggplot(m, aes(V1, V2, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = lab), size = 4) +
  scale_fill_gradient2(low = "#B03A2E", mid = "white",
                       high = "#1F4E79", limits = c(-1, 1)) +
  labs(title = "Correlogram of the Fitness Centre Case (n = 14)", x = NULL, y = NULL)
```

## Output: 7 x 7 correlation matrix

| | outdoor_temp | gym_visits | shakes_sold | load_dev | injuries | trainers | towel_loads |
|---|---|---|---|---|---|---|---|
| **outdoor_temp** | 1.00 | 0.98 | -0.86 | -0.29 | -0.32 | 0.27 | -0.19 |
| **gym_visits** | 0.98 | 1.00 | -0.84 | -0.27 | -0.34 | 0.24 | -0.24 |
| **shakes_sold** | -0.86 | -0.84 | 1.00 | 0.22 | 0.30 | -0.04 | 0.46 |
| **load_dev** | -0.29 | -0.27 | 0.22 | 1.00 | 0.03 | 0.42 | 0.28 |
| **injuries** | -0.32 | -0.34 | 0.30 | 0.03 | 1.00 | -0.51 | 0.05 |
| **trainers** | 0.27 | 0.24 | -0.04 | 0.42 | -0.51 | 1.00 | 0.63 |
| **towel_loads** | -0.19 | -0.24 | 0.46 | 0.28 | 0.05 | 0.63 | 1.00 |

*Cells shaded by strength: strong (|r| > 0.7) darker, moderate (0.3 to 0.7) lighter, weak unshaded. Blue = positive, red = negative.*

**n =** 14 daily records.

**Distinct pairs of variables =** 7 × 6 / 2 = 21 (the 7 diagonal values are each variable with itself and the lower triangle repeats the upper triangle).

## Correlogram

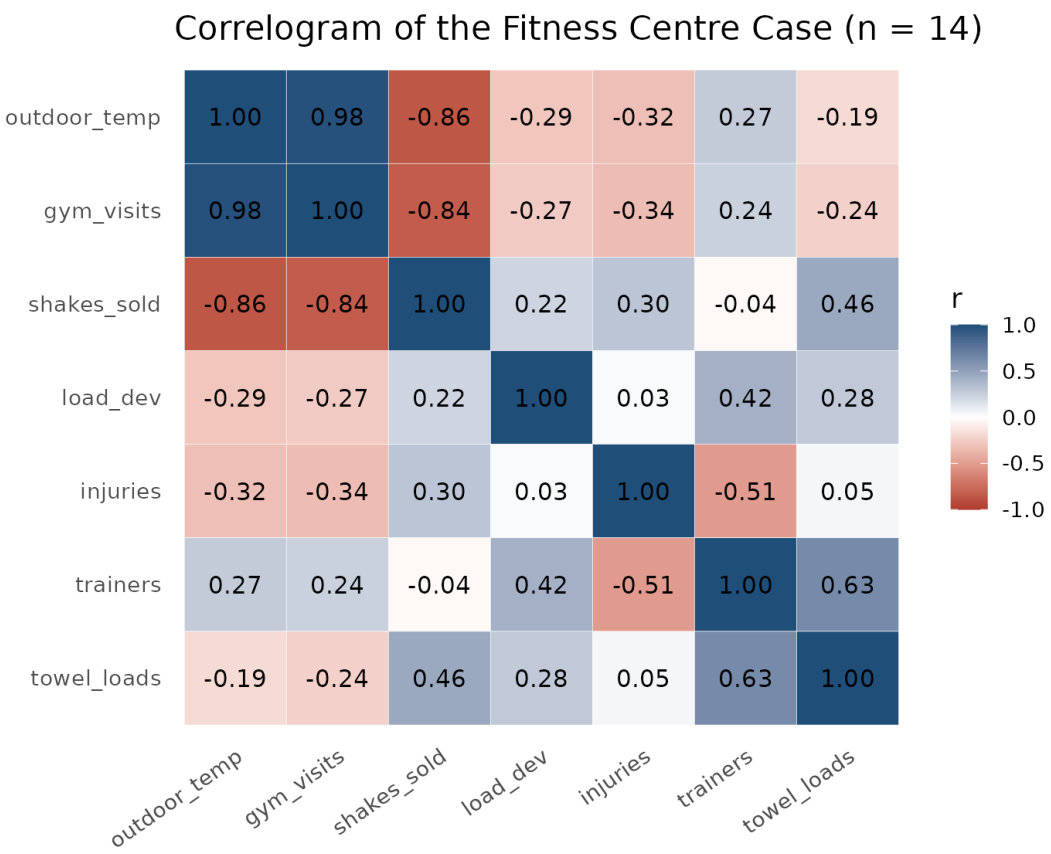



*Figure 1: Correlogram of the 7 variables (ggplot2 geom_tile with geom_text).*

**2.Question:** *Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7).*

### R code

```
cm <- cor(case)
idx <- which(upper.tri(cm), arr.ind = TRUE)
pairs <- data.frame(a = rownames(cm)[idx[, 1]],
                    b = colnames(cm)[idx[, 2]],
                    r = round(cm[idx], 2))
pairs[order(-abs(pairs$r)), ]      # sorted by strength
pairs[which.max(pairs$r), ]         # strongest positive
pairs[which.min(pairs$r), ]         # strongest negative
```

### Answer

| | Pair | r | Strength |
|---|---|---|---|
| **Strongest positive** | outdoor_temp and gym_visits | 0.98 | Strong (> 0.7) |
| **Strongest negative** | outdoor_temp and shakes_sold | -0.86 | Strong (> 0.7) |

Hotter days bring more gym visits (r = 0.98) and fewer protein shakes sold (r = -0.86). The next strongest pair is gym_visits and shakes_sold at -0.84, which is investigated in Task 5.

**3.Question:** *Plot injuries against load_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship.*

### R code

```
r3 <- cor(case$load_dev, case$injuries); round(r3, 2)
fit <- lm(injuries ~ load_dev + I(load_dev^2), data = case)
round(summary(fit)$r.squared, 2)

ggplot(case, aes(load_dev, injuries)) +
  geom_point(size = 3, colour = "#1F4E79") +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
              colour = "#B03A2E", linetype = "dashed") +
  geom_smooth(method = "lm", formula = y ~ x + I(x^2), se = FALSE,
              colour = "#1E8449") +
  labs(title = "Training load deviation vs injuries",
       x = "load_dev (units)", y = "injuries")
```

### Plot

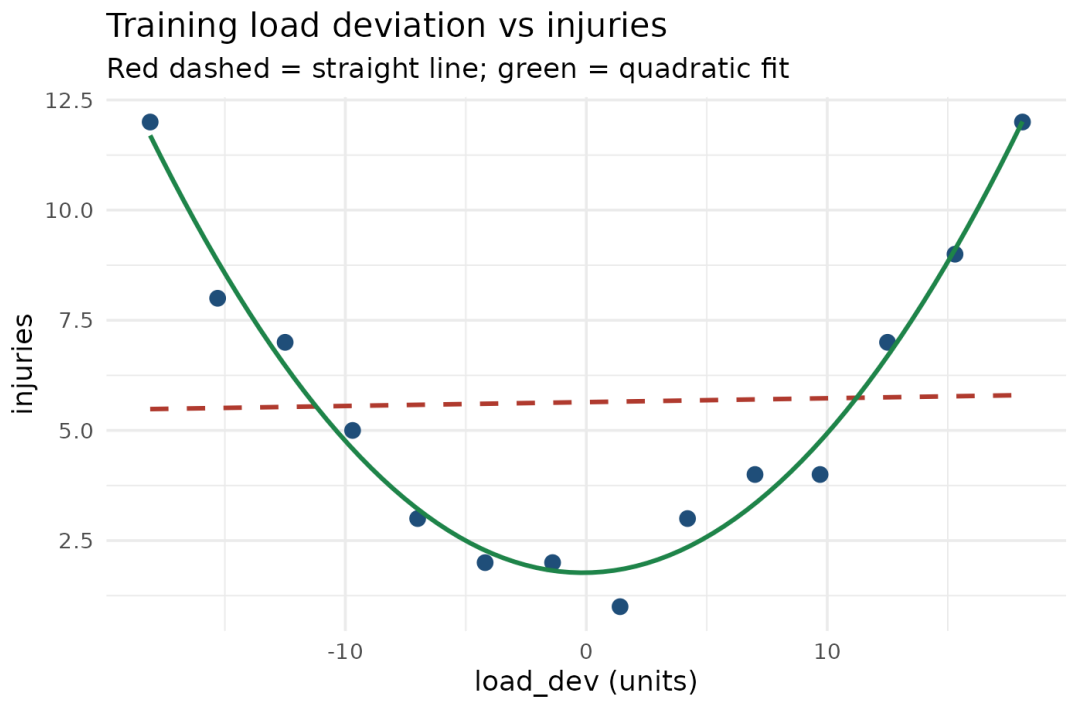



*Figure 2: injuries vs load_dev. The straight line (red, dashed) is flat; the quadratic curve (green) follows the points closely.*

### Answer

| Measure | Value |
|---|---|
| **Pearson r (load_dev, injuries)** | 0.03 (weak, essentially zero) |
| **R-squared of quadratic fit** | 0.98 |

**Why r misses it:** Injuries rise when the training load is far from the optimum in either direction (a U-shape), so the high-injury points on the left and right cancel out and r, which only measures a straight-line link, comes out near zero even though the quadratic curve explains about 98% of the variation.

**4.Question:** *Plot towel_loads against trainers. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r.*

### R code

```
case$flag <- ifelse(seq_len(nrow(case)) == 6, "Row 6", "Other")

ggplot(case, aes(trainers, towel_loads, colour = flag)) +
  geom_point(size = 3.2) +
  geom_smooth(aes(group = 1), method = "lm", formula = y ~ x,
              se = FALSE, colour = "grey40", linetype = "dashed") +
  scale_colour_manual(values = c(Other = "#1F4E79", "Row 6" = "#B03A2E")) +
  labs(title = "Trainers on duty vs towel loads",
       x = "trainers", y = "towel_loads", colour = NULL)

case[6, ]                                              # the suspect row
round(cor(case$trainers, case$towel_loads), 2)         # all rows
round(cor(case$trainers[-6], case$towel_loads[-6]), 2) # without row 6
```

### Plot

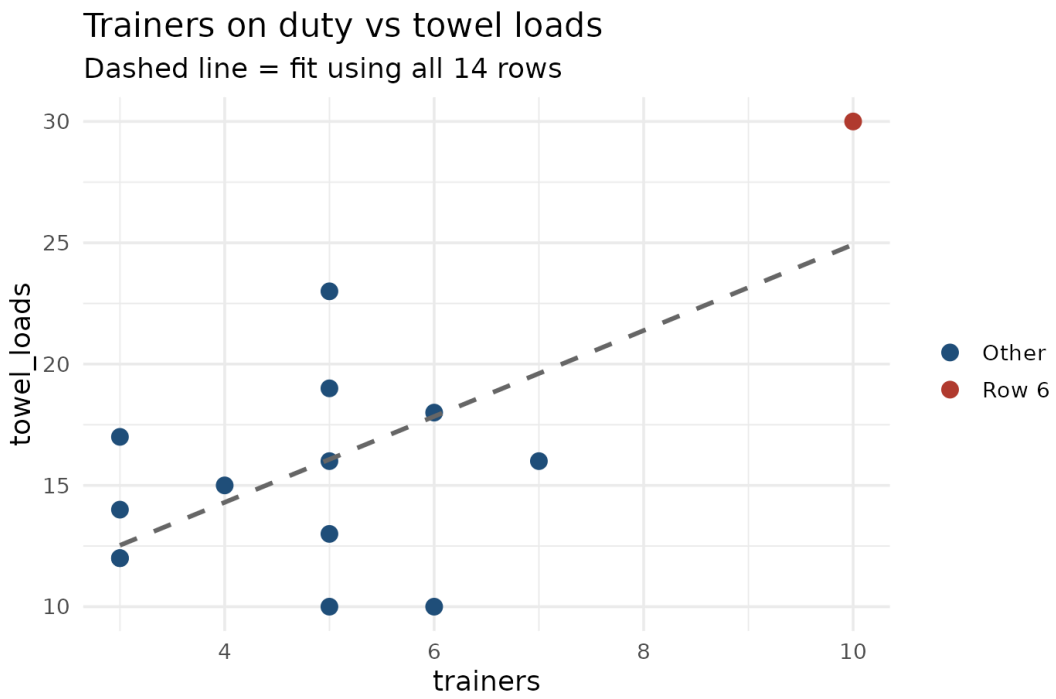



*Figure 3: towel_loads vs trainers. Row 6 (red) sits far from the other 13 points.*

### Answer

| Measure | Value |
|---|---|
| **Suspect row** | Row 6 (trainers = 10, towel_loads = 30) |
| **r with all 14 rows** | 0.63 (moderate) |
| **r without row 6** | 0.17 (weak) |

**What the single record did:** Row 6, the trainee's typed-in entry, inflated r from a weak 0.17 to a moderate 0.63, so one extreme point manufactured most of the apparent link between trainers on duty and towel laundry.

**5.Question:** *gym_visits and shakes_sold are strongly linked. Report r(gym_visits, shakes_sold), then r(outdoor_temp, gym_visits) and r(outdoor_temp, shakes_sold). Compute the partial correlation of gym_visits and shakes_sold controlling for outdoor_temp. Write a two-sentence verdict: does gym_visits cause shakes_sold? Name one study design that could test causation.*

### R code

```
round(cor(case$gym_visits,   case$shakes_sold), 2)   # r(gym_visits, shakes_sold)
round(cor(case$outdoor_temp, case$gym_visits), 2)   # r(outdoor_temp, gym_visits)
round(cor(case$outdoor_temp, case$shakes_sold), 2)  # r(outdoor_temp, shakes_sold)

# Partial correlation controlling for outdoor_temp
res_gym    <- resid(lm(gym_visits  ~ outdoor_temp, data = case))
res_shakes <- resid(lm(shakes_sold ~ outdoor_temp, data = case))
round(cor(res_gym, res_shakes), 2)
```

## Output

| Quantity | Value |
|---|---|
| **r(gym_visits, shakes_sold)** | -0.84 |
| **r(outdoor_temp, gym_visits)** | 0.98 |
| **r(outdoor_temp, shakes_sold)** | -0.86 |
| **Partial r (gym_visits, shakes_sold \| outdoor_temp)** | 0.10 |

## Verdict

No, gym_visits does not cause shakes_sold: outdoor temperature is a hidden driver (a confounder) that pushes gym visits up (r = 0.98) and shake sales down (r = -0.86), and once temperature is controlled the link collapses from r = -0.84 to a partial r of just 0.10. The strong correlation is therefore a by-product of the weather, and correlation alone cannot prove cause.

**Study design to test causation:** A randomised controlled experiment, for example randomly assigning days (or members) to receive a free-entry or promotion that increases gym visits regardless of the weather, then comparing shake sales between the randomised groups while temperature is balanced by the randomisation.